ICAIF 2026 · Milan

Trustworthy Document Intelligence in Finance

A hands-on tutorial · ICAIF 2026

docai.finhorizons.org



Yaxuan Kong
University of Oxford
Organizer · Presenter



Yichen Li
FinHorizons Foundation
Organizer · Presenter



Stefan Zohren
University of Oxford
Organizer

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08:30–10:30 CET

Room 3, Bocconi University
Via Röntgen 1, Milan



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In finance, a good answer is not yet a trustworthy answer.

A good answer reads well and cites sources, while a trustworthy one can be checked number by number.

A GOOD ANSWER

What was NVIDIA's Data Center revenue in fiscal 2026, and how much did it grow?

CLAUDE OPUS 5.5 · 12 NOV 2026 · FIRST PARAGRAPH

Searched the web ›

See the full answer

NVIDIA's Data Center revenue for fiscal 2026 was a record **$193.7 billion, up 68% year over year**. Full-year Data Center revenue rose 68% to $193.7 billion. Working back from that growth rate, fiscal 2025 was about $115 billion, so the segment added roughly $78 billion in one year. hyper

Sources: NVIDIA Investor Relations ↗ Seeking Alpha ↗ HyperAI ↗ VectorShift ↗

WHAT IT STILL NEEDS

1 Proof

The headline number cites a third-party summary, not NVIDIA's filing.

2 Context

The 2025 figure is worked back from a growth rate, not read from a table.

3 Date

Ask about a past date, and it uses documents published after that date.

4 Record

Nothing is saved, so the same question can give another answer tomorrow.

A TRUSTWORTHY ANSWER click a number

What was NVIDIA's Data Center revenue in fiscal 2026, and how much did it grow?

NVIDIA's Data Center revenue was **$193.7 billion** in fiscal 2026, up **68%** from **$115.2 billion** in fiscal 2025.

1 2 10-K · Note 16 · Revenue by End Market

(In millions)	Jan 25, 2026	Jan 26, 2025	Jan 28, 2024
Data Center	**$193,737**	**$115,186**	**$47,525**
Compute	162,361	102,196	38,950
Networking	31,376	12,990	8,575

See it in the 10-K

3 The 10-K was public on 25 February 2026.

4 The answer and its sources are kept on record.

Data security requires financial institutions to build their own systems, so trustworthiness should be considered from the start.

An example of a trustworthy QA system for financial documents

PREPARE THE DOCUMENTS: Documents › Parse › Chunk › Index → ANSWER A QUESTION: Question › Retrieve › Generate › Numeric grounding › Answer › Audit trail → KEEP IMPROVING: Evaluate › Improve → TOGETHER: Whole system

Documents

STEP 1 OF 12 · PREPARE THE DOCUMENTS

In practice, a firm holds the filings of many companies across all industries, and each company regularly adds new filings.

THE FILINGS A FIRM HOLDS · CLICK A COMPANY

SEMICONDUCTORS | BANKS | ENERGY | HEALTHCARE

For every company the system takes the annual report (10-K), the quarterly reports (10-Q) and the results announcements (8-K), and it can take the firm's own notes too.

▶ Play the story

Click any step above, or let the story play by itself.

Next: Parse →

Today’s agenda

08:30 **Opening**
10 min · Yaxuan

08:40 **Block 1 · Parsing, chunking and indexing**
25 min · Yaxuan · exercise 1

09:05 **Block 2 · Retrieval, generation, numeric grounding**
30 min · Yaxuan · exercise 2

09:35 **Ask the documents · live demo**
5 min · any question, from your phone

09:40 **Block 3 · Faithfulness evaluation**
25 min · Yaxuan · exercise 3

10:05 **Block 4 · Production concerns**
15 min · Yichen

10:20 **Wrap-up and your questions**
10 min · both · ends 10:30 CET

What we provide

click a card for details

The notebook

Open in Colab ↗

```
ground_numbers(answer)
$215.9 billion      VERBATIM
$130,387 million    VERBATIM
60.4%               DERIVED
55,000              UNGROUNDED
```

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4 NVIDIA filings ›

- **10-K** fiscal 2026
- **Press release** Q4 fiscal 2026
- **10-Q** Q2 fiscal 2027
- **CFO commentary** Q4 fiscal 2026

15 test questions ›

What was NVIDIA’s GAAP operating margin for fiscal year 2026?

Gold answer: **about 60.4%**

Ask the documents ›

In fiscal year 2026 NVIDIA repurchased **282 million** shares for **$40.4 billion** [1].

docai.finhorizons.org/ask

250 checks ›

✕ A margin of “4.9%” passed, because the cited table held an EPS of “$4.90”.

✓ Now the kinds must agree.

800+ papers and tools ›

Parse 133 · Chunk 40 · Index 115 · Retrieve 72 · Generate 95 · Numbers 48 · Evaluate 230

The slides ›



Web page · PowerPoint · PDF

The data behind every example today

The system reads four public filings of NVIDIA, and two more files are used to check what it answers.

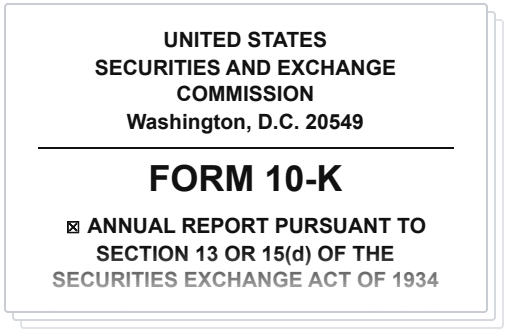
WHAT THE SYSTEM READS

click a filing to open it

25 February 2026
NVIDIA reports fiscal 2026, the year that ended on 25 January 2026.

six months later

26 August 2026
Fiscal 2027's second quarter follows.


UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

Form 10-K · annual report

filed 25 Feb 2026

year ended 25 Jan 2026

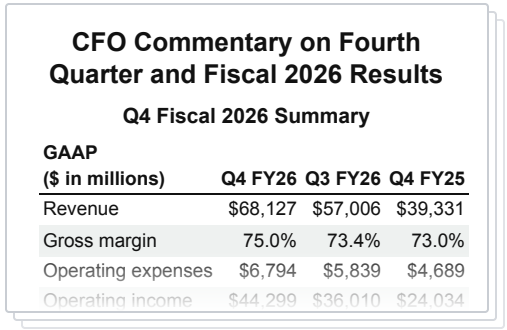

NVIDIA Announces Financial Results for Fourth Quarter and Fiscal 2026

- Record quarterly revenue of $68.1 billion, up 20% from Q3 and up 73% from a year ago
- Record quarterly Data Center revenue of $62.3 billion, up 22% from Q3 and up 75% from a year ago

Press release

filed 25 Feb 2026

8-K exhibit 99.1


CFO Commentary on Fourth Quarter and Fiscal 2026 Results

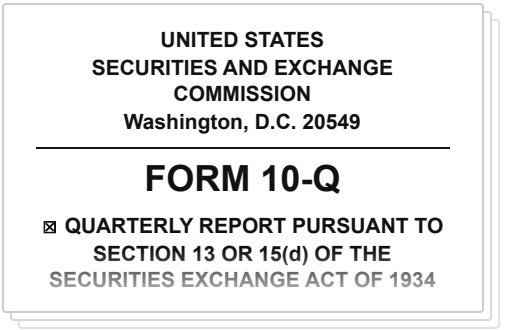
Q4 Fiscal 2026 Summary

GAAP ($ in millions)	Q4 FY26	Q3 FY26	Q4 FY25
Revenue	$68,127	$57,006	$39,331
Gross margin	75.0%	73.4%	73.0%
Operating expenses	$6,794	$5,839	$4,689
Operating income	$44,299	$36,010	$24,034

CFO commentary

filed 25 Feb 2026

8-K exhibit 99.2


UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-Q

☒ QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

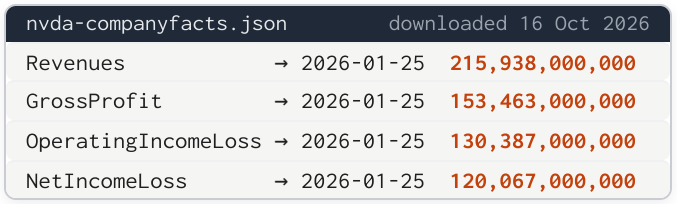
Form 10-Q · quarterly report

filed 26 Aug 2026

quarter ended 26 Jul 2026

WHAT CHECKS ITS ANSWERS

click a file to look inside

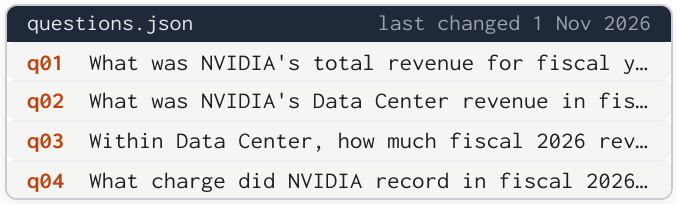


nvda-companyfacts.json — downloaded 16 Oct 2026

Revenues	→	2026-01-25	215,938,000,000
GrossProfit	→	2026-01-25	153,463,000,000
OperatingIncomeLoss	→	2026-01-25	130,387,000,000
NetIncomeLoss	→	2026-01-25	120,067,000,000

NVIDIA's numbers as data

The SEC publishes every number that NVIDIA reports.



questions.json — last changed 1 Nov 2026

q01	What was NVIDIA's total revenue for fiscal y…
q02	What was NVIDIA's Data Center revenue in fis…
q03	Within Data Center, how much fiscal 2026 rev…
q04	What charge did NVIDIA record in fiscal 2026…

15 evaluation questions

Each one has an answer that a person checked.

Run this in Colab ↗

Section 0 of the notebook downloads these files and loads them.

Part 1: Parsing, chunking and indexing

- **Parsing** turns a filing, in whatever form it arrives, into text and tables that a machine can read.
- **Chunking** cuts each parsed filing into small pieces that can be searched, and it copies the filing's metadata onto every piece.
- **Indexing** stores the pieces in catalogues, so that a search finds the right

What is parsing?

A filing can arrive in many forms. Parsing turns each of them into text and tables that a machine can read.

THE SAME LINES OF NVIDIA'S 10-K, THREE TIMES

Item 7 · Fiscal Year 2026 Summary

WHAT A PERSON SEES

Fiscal Year 2026 Summary

($ in millions)	Jan 25, 2026	Jan 26, 2025	Change
Revenue	$215,938	$130,497	Up 65%
Gross margin	71.1%	75.0%	-3.9 pts
Operating expenses	$23,076	$16,405	Up 41%
Operating income	$130,387	$81,453	Up 60%
Net income	$120,067	$72,880	Up 65%

Revenue for fiscal year 2026 was $215.9 billion, up 65% from a year ago.

A person can tell by eye that the bold line is a heading, and that $215,938 is the revenue of the year that ended in January 2026.

WHAT THE FILE HOLDS · attributes trimmed

nvda-10k-fy2026.htm

```
<span style="font-weight:700">
  Fiscal Year 2026 Summary</span>
<table>
 <tr><td>Revenue</td><td>$</td>
     <td>215,938</td><td>$</td>
     <td>130,497</td><td>Up 65%</td></tr>
 …
</table>
<span>Revenue for fiscal year 2026 was
  $215.9 billion, up 65% from a year ago.</span>
```

The file does not say any of this. It only marks one line as bold, and it keeps each "$" in a different cell from its number.

WHAT PARSING RETURNS · the two records in full

TWO RECORDS · JSON — scroll to see all of it

TABLE

```
{ "table_id": "nvda-10k-fy2026:t010",
  "doc_id": "nvda-10k-fy2026",
  "item": "Item 7",
  "section_title": "Fiscal Year 2026 Summary",
  "unit_text": "in millions, except per share data",
  "scale": 1000000.0, "header_rows": 3,
  "rows": [ ["Year Ended"],
    ["Jan 25, 2026", "Jan 26, 2025", "Change"],
    ["($ in millions, except per share data)"],
    ["Revenue", "$215,938", "$130,497", "Up 65%"],
    ["Gross margin", "71.1%", "75.0%", "-3.9 pts"],
    ["Operating expenses", "$23,076", "$16,405", "Up
```

Parsing writes it down: this block is a table, its title is "Fiscal Year 2026 Summary", it is in Item 7, and its numbers are in millions.

THE FORM IN WHICH A FILING ARRIVES DECIDES HOW IT IS READ

click a card or a tool to open it

HTML A web page

The parser reads the tags of the page. The filings on EDGAR arrive in this form.

Tools: BeautifulSoup, lxml, Unstructured

PDF A digital PDF

The parser reads each word with its position on the page, and it rebuilds lines, columns and tables.

Tools: pdfplumber, PyMuPDF, Docling

IMAGE A scan or a photo

A model reads the characters from the picture, so every number is a transcription that still needs a check.

Tools: Tesseract, PaddleOCR, vision models

XLSX A spreadsheet

The parser reads the cells. It still has to find the header row, the unit and the end of each table.

Tools: openpyxl, pandas, DuckDB (SQL)

Run this in Colab ↗

Section 1.1 of the notebook parses NVIDIA's 10-K into 23 items, 1,002 paragraphs and 64 tables.

What can go wrong in parsing

Six parts of a filing are easy for a parser to get wrong. Each example below is taken from NVIDIA's 10-K.

1 **Tables** — A row loses its header
2 **Units and signs** — Millions, brackets and dashes
3 **Pictures and scans** — A number that is only an image
4 **Page breaks** — Page numbers inside the text
5 **Headings** — The contents page repeats them
6 **Footnotes** — A note far from its table

A table read as plain text loses the header row that says which year each number belongs to.

IN THE 10-K

Consolidated Statements of Income
(In millions, except per share data)

	Jan 25, 2026	Jan 26, 2025	Jan 28, 2024
Revenue	$215,938	$130,497	$60,922
Cost of revenue	62,475	32,639	16,621
Gross profit	153,463	97,858	44,301
Operating income	130,387	81,453	32,972
Net income	$120,067	$72,880	$29,760

A CARELESS PARSER RETURNS

```
Revenue $215,938 $130,497 $60,922
Cost of revenue 62,475 32,639
16,621 Gross profit 153,463 97,858
44,301 … Operating income 130,387
81,453 32,972 … Net income
$120,067 $72,880 $29,760
```

✕ The header row is gone, so nothing says which year 130,497 belongs to.

A CAREFUL PARSER KEEPS

```
table  Statements of Income
unit   in millions
head   Jan 25, 2026|Jan 26, 2025
row    Revenue|215,938|130,497
row    Net income|120,067|72,880
```

✓ The number keeps its row, its column and its unit, so a citation can name the cell.

WHY IT MATTERS — Most of the numbers that analysts ask about stand in tables.

WHAT TO DO — Keep each table as one block with its header row, its unit line and its row labels.

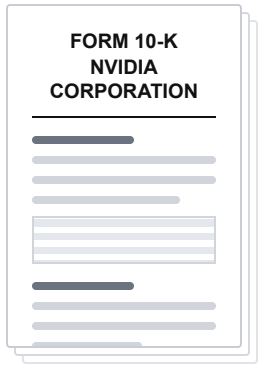
Run this in Colab ↗ Section 1.1 of the notebook runs our parser on the full 10-K: 64 tables kept intact and every skipped picture listed.

What is chunking?

Chunking cuts each parsed filing into small pieces that can be searched, and it copies the filing's metadata onto every piece.

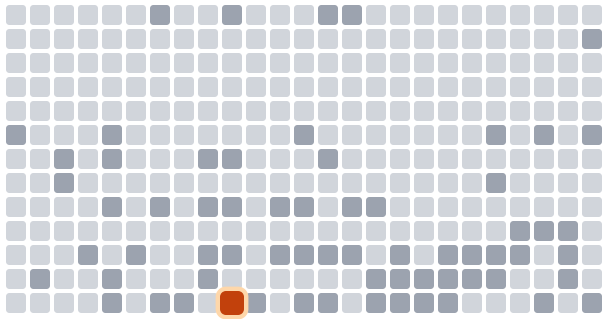
FROM ONE FILING TO PIECES THAT CAN BE SEARCHED

ONE FILING



NVIDIA's 10-K has about 50,000 words.

325 PIECES · each square is one piece



259 pieces are groups of paragraphs and 66 are tables. **Click any square** to read that piece.

ONE PIECE · its text in Markdown, and its metadata

text · Markdown — piece 310 of 325 · 144 words

```
[10-K FY2026 › Item 15 › The following table summarizes revenue by
specialized markets: (In millions)]
Rows: Data Center; Compute; Networking; Gaming; Professional
Visualization; Automotive; OEM and Other; Total revenue
| Year Ended |  |  |  |
|  | Jan 25, 2026 | Jan 26, 2025 | Jan 28, 2024 |
| Revenue by End Market: | (In millions) | (In millions) | (In millions) |
|---|---|---|---|
|  Data Center | $193,737 | $115,186 | $47,525 |
|  Compute | 162,361 | 102,196 | 38,950 |
|  Networking | 31,376 | 12,990 | 8,575 |
|  Gaming | 16,042 | 11,350 | 10,447 |
```

scroll to see all of it

metadata

copied from the filing · the same on every piece

doc_id	nvda-10k-fy2026
filing_date	2026-02-25
period_end	2026-01-25

of this piece only

chunk_id	847622ce5b02
chunk_type	table
item	Item 15
table_id	t056
scale	1,000,000

The text is what a search reads. The metadata is what a citation, a date filter and an audit trail use. Every piece carries both: in another square only the text and the piece's own fields change.

WHY A FILING IS CUT

Search needs pieces, as searching a whole 10-K would match every question and would not show where the answer is.

The model needs pieces, as it cannot read 50,000 words for every question. It reads the six pieces that search returns.

A citation needs pieces, as a person has to check it. One piece is 144 words to read, and a whole 10-K is 50,000.

HOW LARGE A PIECE SHOULD BE

the question asks for Data Center revenue in fiscal year 2026

Too small one row	✗	`| Data Center | $193,737 | $115,186 | $47,525 |` has lost its header, so nothing says which number is fiscal year 2026, or that the unit is millions.
About right one table	✓	The whole table is one piece of 144 words. Its years and its unit stay with every row, so $193,737 million can be read and checked.
Too large all of Item 15	✗	The number is somewhere in 21,600 words. The model has to read all of them, and the citation no longer points at a place.

Run this in Colab ↗

Section 1.3 of the notebook cuts three filings into 383 pieces: 325 from the 10-K, 32 from the press release and 26 from the CFO commentary.

Eight ways to cut a filing into pieces

Each strategy cuts the same filing in a different place. A good cut keeps a table whole, and a bad one splits it in two.

WHERE TO CUT · BY SIZE

Fixed size ●
A cut every 400 words

Sentences and paragraphs ●
A cut where a paragraph ends

Pages
One page is one piece

WHERE TO CUT · BY MEANING

Structure-aware ●
A cut where a section ends

Semantic
A cut where the subject changes

WHAT TO ADD · TO EACH PIECE

Parent and child
Search small, read large

Context line ●
Each piece says where it is from

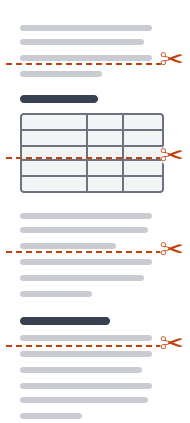
Model-written statements
One fact in each piece

● runs in our notebook

Fixed size

RUNS IN OUR NOTEBOOK · REAL PIECES

A piece ends at its 400th word, even inside a table, and its last 50 words are repeated in the next piece.



piece 103 of 145 · its last 90 words

… conditions involved evaluating whether the assumptions used by management were reasonable considering (i) current and past results, including historical product life cycle, (ii) the consistency with external market and industry data, and (iii) changes in technology. /s/ PricewaterhouseCoopers LLP San Jose, California February 25, 2026 We have served as the Company's auditor since 2004. 50 NVIDIA Corporation and Subsidiaries Consolidated Statements of Income (In millions, except per share data) Year Ended Jan 25, 2026 Jan 26, 2025 Jan 28, 2024 Revenue $215,938 $130,497 $60,922 Cost of revenue 62,475 32,639 16,621

piece 104 · its first 95 words

Jose, California February 25, 2026 We have served as the Company's auditor since 2004. 50 NVIDIA Corporation and Subsidiaries Consolidated Statements of Income (In millions, except per share data) Year Ended Jan 25, 2026 Jan 26, 2025 Jan 28, 2024 Revenue $215,938 $130,497 $60,922 Cost of revenue 62,475 32,639 16,621 Gross profit 153,463 97,858 44,301 Operating expenses Research and development 18,497 12,914 8,675 Sales, general and administrative 4,579 3,491 2,654 Total operating expenses 23,076 16,405 11,329 Operating income 130,387 81,453 32,972 Interest income 2,300 1,786 866 Interest expense (259) (247) (257) Other income, net 9,022 …

The words in orange are the 50 that both pieces hold. Piece 103 ends after the second row of the income statement, so the table is cut in two.

ADVANTAGE

It is the simplest strategy, and every piece has the same size, so cost and speed are easy to plan.

DISADVANTAGE

A cut can fall inside a sentence or a table, and a piece does not know which section it comes from.

WHEN TO CHOOSE IT

Choose it for plain text without structure, and as the baseline that every other strategy has to beat.

Run this in Colab ↗

Section 1.3 of the notebook cuts the 10-K three ways: into 145 pieces of fixed size, 172 paragraph pieces and 325 structure-aware pieces.

What chunking strategy to choose

Choose by what your documents look like, then test the choice on your own questions.

1 START FROM WHAT YOU HAVE — click one

YOUR DOCUMENTS

- **Filings with tables**
- News and research notes
- Slide decks and scans
- Call transcripts
- Any text, as a baseline

YOUR QUESTIONS

- Many similar documents
- A detail in a long section
- Single facts, no numbers

CUT IT THIS WAY

Structure-aware

RUNS IN OUR NOTEBOOK



A cut where a section ends

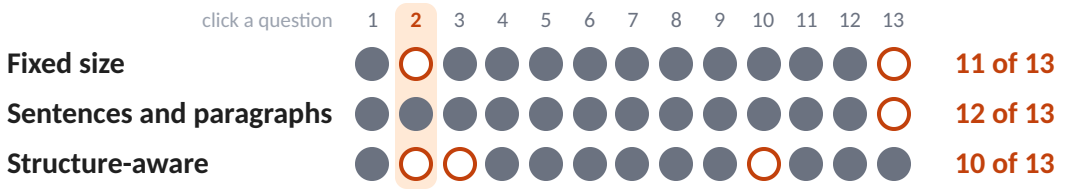
A table stays whole and a piece knows its section.

2 TEST THE CHOICE ON YOUR OWN QUESTIONS — what our notebook measured

We ask each of our 13 number questions and read the first 1,500 words that the search returns.

QUESTION 2 its numbers stand only in tables

What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?



No clear winner | **Tables are harder to find** | **Found, but without its columns**

The three strategies find the right numbers for 11, 12 and 10 of the 13 questions. A gap that small does not show which one is best.

	the first 1,500 words	the first 3,000 words
Fixed size	**11**	**11**
Sentences and paragraphs	**12**	**12**
Structure-aware	**10**	**11**

Run this in Colab ↗ Section 2.2 of the notebook runs this test on the three chunking strategies and lists the questions that each one missed.

What is indexing?

Indexing stores the pieces in catalogues, so that a search finds the right pieces without reading every filing again.

A SEARCH READS THE CATALOGUE, NOT THE FILINGS

click a word or a question

1 · CHOOSE WHAT TO LOOK UP

A catalogue by words the keyword index

For every word, it notes which pieces contain it, like the index at the back of a book.

revenue · Blackwell · H20 · repurchased · **193,737** · buybacks

A catalogue by meaning the vector index

A model places every piece on a map of meanings. Pieces about the same subject lie close together, whatever words they use.

"What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?"

"How much did NVIDIA spend repurchasing shares in fiscal 2026 …?"

2 · INSIDE THE CATALOGUE

the catalogue by words — 5,309 words, A to Z

1933	4 pieces
1934	9 pieces
193479	3 pieces
193737	2 pieces → 310, 368
1940	1 piece
195	1 piece
1959	1 piece

An exact number is found at once, and one of the two pieces is the Note 16 table.

The catalogue lists every word of the 383 pieces once. A search reads this one line, and not the pieces.

3 · THE PIECES IT NAMES · one square is one piece



193,737 → 2 pieces of 383

10-K › Item 15 › The following table summari… the Note 16 table · piece 310 · table

… | | | | | Jan 25, 2026 | Jan 26, 2025 | Jan 28, 2024 | | Revenue by End Market: | (In millions) | (In millions) | (In millions) | |---|---|---|---| | Data Center | $ **193,737** | $115,186 | $47,525 | | Compute | 162,361 | 102,196 | 38,950 | | Networking | 31,376 | 12,990 | 8,575 | | Gaming | 16,042 | 11,350 | 10,447 | | Professional…

CFO commentary › Fiscal 2026 Summary › Revenue by Market Pl… piece 368 · table

… Professional Visualization; Automotive; OEM and Other; Total | Revenue by Market Platform | | | | ($ in millions) | FY26 | FY25 | Y/Y | |---|---|---|---| | Data Center | $ **193,737** | $115,186 | 68% | | Compute | 162,361 | 102,196 | 59% | | Networking | 31,376 | 12,990 | 142% | | Gaming | 16,042 | 11,350 | 41% | | Professional…

WHEN THE CATALOGUES ARE WRITTEN, AND WHAT IS KEPT WITH EVERY PIECE

Written once. The catalogues are written when a filing arrives, before any question is asked, so a search does not have to read the 383 pieces again.

Labels on every piece. Each piece keeps its metadata from chunking, so a search can first set aside the filings that did not yet exist on a given day.

A new filing. Six months later the 10-Q arrives. Its 181 pieces are added to the catalogues, and the 383 earlier pieces stay as they are.

Run this in Colab ↗ Section 2.0 of the notebook builds two catalogues for the 383 pieces: one by words (the keyword index) and one by meaning (the vector index).

Eight ways to index a filing

No single index finds every question, so a system combines several of them.

FIND · BY WORDS

Keyword index ●
Each word points to its pieces

Learned sparse index
A model adds related words

FIND · BY MEANING

Vector index ●
Each piece is a point on a map

Multi-vector index
Each word is a point of its own

COMBINE · AND FILTER

Hybrid index ●
Words and meaning, merged

Metadata index ●
Labels filter the pieces first

ADD A LAYER · WRITTEN BY A MODEL

Summary tree
Summaries above the pieces

Knowledge graph
Entities and their links

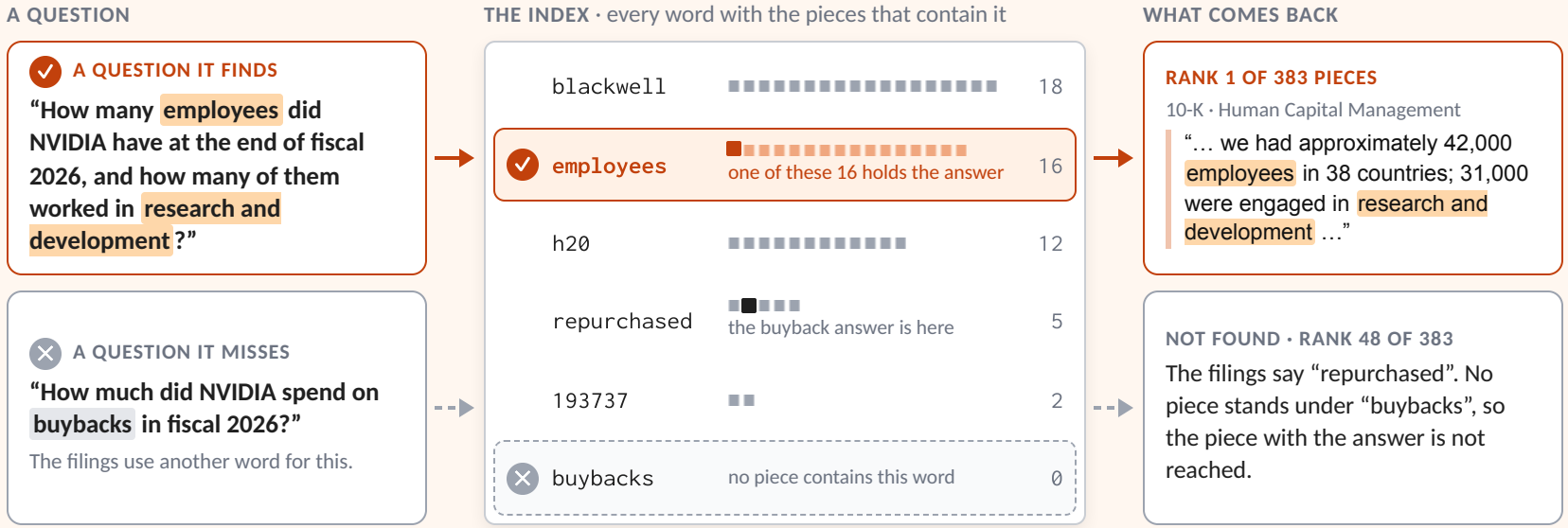
● runs in our notebook

Keyword index

RUNS IN OUR NOTEBOOK · REAL COUNTS AND RANKS

usual scoring rule: BM25 ?

Every word is listed with the pieces that contain it, and a rare word counts for more than a common one.



ADVANTAGE

Exact names, codes and numbers are found as they are written, and no model is needed.

DISADVANTAGE

It finds only the words of the question, so another wording of the same thing is missed.

WHEN TO CHOOSE IT

Choose it whenever questions contain product names, item numbers or figures.

Run this in Colab ↗

Section 2.0 of the notebook searches one question three ways: by words, by meaning and with both merged.

Exercise 1 · Parse, chunk and index the 10-K

Your task is to do each step yourself on one table of the 10-K, the table that answers our question about Data Center revenue.

1 **Parse** — From the file to a table ▸ 2 **Chunk** — From the filing to pieces ▸ 3 **Index** — From a piece to the catalogue

This is what the file holds for one table. **Click the four things that a parser has to find in it**: the title, the years, the unit and the Data Center row. show all four

nvda-10k-fy2026.htm · what the file holds — attributes and empty cells trimmed

```
<div>The following table summarizes revenue by specialized markets:</div>
<table>
 <tr><td>Year Ended</td></tr>
 <tr><td>Jan 25, 2026</td><td>Jan 26, 2025</td><td>Jan 28, 2024</td></tr>
 <tr><td>Revenue by End Market:</td><td>(In millions)</td></tr>
 <tr><td>Data Center</td><td>$</td><td>193,737</td>
     <td>$</td><td>115,186</td><td>$</td><td>47,525</td></tr>
 <tr><td>Compute</td><td>162,361</td><td>102,196</td><td>38,950</td></tr>
 <tr><td>Networking</td><td>31,376</td><td>12,990</td><td>8,575</td></tr>
 <tr><td>Gaming</td><td>16,042</td><td>11,350</td><td>10,447</td></tr>
 <tr><td>Professional Visualization</td><td>3,191</td><td>1,878</td><td>1,553</td></tr>
 <tr><td>Automotive</td><td>2,349</td><td>1,694</td><td>1,091</td></tr>
 <tr><td>OEM and Other</td><td>619</td><td>389</td><td>306</td></tr>
 <tr><td>Total revenue</td><td>$</td><td>215,938</td>
     <td>$</td><td>130,497</td><td>$</td><td>60,922</td></tr>
</table>
```

WHAT PARSING RETURNS · one table that a machine can read

Title	click it in the file
Years	click it in the file
Unit	click it in the file
Row	click it in the file

Find all four, and the table can be read without the file.

Run this in Colab ↗

Exercise 1 of the notebook finds the three financial statements in the parsed 10-K and checks that total revenue for fiscal year 2026 is 215,938.

Part 2: Retrieval, generation and numeric grounding

- **Retrieval** chooses the few pieces that the model will read.
- **Generation** writes the answer from these pieces.
- **Numeric grounding** checks every number of the answer against its source.
- **How much machinery** a system needs depends on the question.

What is retrieval?

Retrieval searches the pieces and selects the few that fit the question best, and the model reads only these.

FROM 564 PIECES TO THE SIX THAT THE MODEL READS

click a step

1 · THE QUESTION

asked on 1 March 2026

What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?

1 Filter by date 564 → **383**
Filings that came out after the day of the question are left out.

2 Search 383 → **40**
The question is looked up in both catalogues, by words and by meaning.

3 Re-rank 40 → **6**
A second model reads every candidate with the question and keeps the best.

2 · THE LIBRARY · one square is one piece

383 of 564 pieces may be read

Form 10-K 325 pieces

Press release 32 pieces

CFO commentary 26 pieces

~~Form 10-Q~~ ~~181 pieces~~ ✕ **left out, filed 26 August 2026**

may be read · left out

The 10-Q did not exist on the day of the question, so its 181 pieces are left out before any search.

3 · WHAT THE STEP GIVES

Step 1 gives 383 pieces to search

The 10-Q was filed on 26 August 2026, after the question was asked. Its 181 pieces are left out.

WHY THE DATE MATTERS · one question, asked on two days

What was NVIDIA's most recently reported quarterly revenue?

as of 1 March 2026	as of 1 September 2026
$68.1 billion	**$96,221 million**
press release, 25 February	10-Q, 26 August

Both answers are right. The date decides which one is given.

THE ADVANTAGE AND THE LIMITATION OF EACH STEP

	1 Filter by date	2 Search	3 Re-rank
ADVANTAGE	is exact: no filing from after the day of the question can reach the model.	is fast: it looks at all 383 pieces in about 30 milliseconds.	is thorough: it reads the text of each candidate and scores how well it answers the question.
LIMITATION	It only allows or forbids a piece, so it cannot say which of the 383 pieces answers the question.	It reads the catalogues and never the pieces, so the right piece can stand far down its list.	It can leave out the right piece: here the two tables with the answer stand 15th and 16th.

Run this in Colab ↗

Sections 2.0 and 2.1 of the notebook run steps 2 and 3: the search returns 40 candidates and the re-ranking keeps 6 of them.

Better retrieval starts with a better question

A better question helps the search find the right piece. The re-ranker then decides whether the model reads it.

SEARCH ONCE

- **As it is** ● — The question is the search
- **Fill in what is missing** — Company, period and form
- **Rewrite it** — The filing's own words
- **A drafted answer** — Search with a guess

SEARCH SEVERAL TIMES

- **Several wordings** — The same question, three ways
- **Sub-questions** — One question becomes several
- **Step back** — First the general question

DO NOT SEARCH YET

- **Ask back** — When the question is unclear

● runs in our notebook

What we found →

As it is

RUNS IN OUR NOTEBOOK · THE RECORDED RUN

THE PROBLEM Our question says "fiscal year 2026" and the table that holds the answer says "Jan 25, 2026". The search ranks twelve pieces of text above the table.

1 · THE QUESTION IS NOT CHANGED

THE QUESTION AS ASKED

What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?

asked directly

≠ the table names the same two years in other words

THE TABLE THAT HOLDS THE ANSWER · 10-K, Note 16, in millions

Year Ended	Jan 25, 2026	Jan 26, 2025
Data Center	$193,737	$115,186

2 · THEN RETRIEVE

the three steps of the last page

1. **Filter by date** 564 → 383
2. **Search** 383 → 40
3. **Re-rank** 40 → 6

3 · WHERE THE ANSWER STANDS

ASKED DIRECTLY

2 **24th** after the search: 13th by words, 45th by meaning

3 **15th** after re-ranking

The model reads six pieces, so it never reads the table.

WHAT IT COSTS

The search depends on how the user happened to word the question.

WHEN TO CHOOSE IT

Choose it as the starting point, and measure what it misses before anything is added.

Run this in Colab ↗

Our notebook searches the question as it is. The other ways were measured with its two models over the same 383 pieces.

What can still go wrong in retrieval

Retrieval can go wrong in six ways. Each example below is taken from the recorded run of our own system.

1 **Ranking** — The right piece is left out
2 **Period** — Which quarter, which year?
3 **Filing date** — A filing from the future
4 **Duplicates** — The same table arrives twice
5 **Conflicts** — Two filings, two numbers
6 **Two sources** — The answer needs two filings

The search ranked the table with the answer 15th and the model reads only the top six pieces.

THE QUESTION · Q02 OF OUR SET

“What was NVIDIA’s Data Center revenue in fiscal year 2026 and in fiscal year 2025?”

THE CORRECT ANSWER

$193,737 million for fiscal 2026 and **$115,186 million** for fiscal 2025, from one table in Note 16 of the 10-K.

RETRIEVAL RANKS THE PIECES · THE MODEL READS THE TOP SIX

1	Press release › NVIDIA Anno…	$62.3 billion · one quarter	text
2	10-K › Note 16 - Segment Information		text
3	10-K › Fiscal Year 2026 Summary	up 68% · no amount	text
4	10-K › Note 9 - Balance Sheet Components		text
5	10-K › Recent Developments, Future Objectives and Ch…		text
6	Press release › Outlook		text

the model reads only the six above this line

7–14	eight more pieces		
15	**10-K › The following table …**	$193,737 million · the year	table

✕ The table with the answer stands at rank 15, below the line, so the model never sees it.

THE MODEL ANSWERS · AS RECORDED

For fiscal year 2026, NVIDIA's Data Center revenue was **$62.3 billion**, up 68% from a year ago. [3] For fiscal year 2025, NVIDIA's Data Center revenue was $38.6 billion, up 54% from a year ago. [3]

✕ **Wrong answer.** $62.3 billion stands in piece 1 and is the revenue of one quarter. Piece 3, cited as [3], holds only “up 68%”.

WHY IT MATTERS — A model cannot quote a table that it never received, however good the model is.

WHAT TO DO — For a question about a number, tables also need priority when the six pieces are chosen.

Run this in Colab ↗ Section 2.2 of the notebook runs retrieval on 13 questions of our set and checks whether the correct numbers are in the pieces it returns.

What is generation?

Generation writes the answer from the six pieces, under rules that make every sentence of the answer checkable.

WHAT THE MODEL IS GIVEN · CLICK A RULE OR A SOURCE

FIVE RULES the instructions, the same for every question

1. **Name the source**
   Every sentence ends with the number of its source, like [3].
2. **Copy the figures**
   A number is written as the source writes it, with its unit and its period.
3. **Show the computation**
   A computed number comes with its division written out.
4. **Say when it is not there**
   If the sources lack the answer, the reply is one fixed sentence.
5. **Keep it short**
   At most four sentences.

SOURCES the six pieces from retrieval, numbered 1 to 6

1 Press release · Q4 results	2 10-K · Note 16	3 10-K · Item 7 summary
4 10-K · Note 9	5 10-K · Developments	6 Press release · Outlook

QUESTION What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?

LLM

WHAT THE MODEL WRITES · THE RECORDED ANSWER · CLICK A CITATION

For fiscal year 2026, NVIDIA's Data Center revenue was $62.3 billion, up 68% from a year ago. [3]

For fiscal year 2025, NVIDIA's Data Center revenue was $38.6 billion, up 54% from a year ago. [3]

The answer is wrong. The filing says $193,737 million for fiscal year 2026. $62.3 billion is the revenue of one quarter.

The answer against the five rules click a rule

1 **Name the source**	✓ both sentences end with [3]
2 **Copy the figures**	✗ $62.3 billion is not in source 3
3 **Show the computation**	– nothing is computed
4 **Say when it is not there**	✗ the sources lack the answer
5 **Keep it short**	✓ two sentences

The rules do not make the model right. They make every sentence checkable, and that is how the wrong number is found.

Run this in Colab ↗

Section 2.4 of the notebook gives the model these five rules and asks it a question. The answers on this page are real outputs of our system.

What is numeric grounding?

Numeric grounding looks for every number of the answer in the sources it cites, and it marks each number that is not there.

1 The answer cites its source

click a number

For fiscal year 2026, NVIDIA's Data Center revenue was **$62.3 billion** ✗, up **68%** ✓ from a year ago. [3]

For fiscal year 2025, NVIDIA's Data Center revenue was **$38.6 billion** ✗, up **54%** ✗ from a year ago. [3]

✗ 1 of 4 numbers has a source, so the answer is flagged.

AN ANSWER THAT COMPUTES · AN ILLUSTRATION

Data Center revenue grew by **$78.6 billion** ✓: 193,737 − 115,186 = 78,551 million. [1]

2 The number is looked for there

$62.3 billion Is it in source [3]?

[3] **10-K · Item 7 · Fiscal Year 2026 Summary**

"Revenue for fiscal year 2026 was $215.9 billion, up 65% from a year ago. Data Center revenue for fiscal year 2026 was up 68% from a year ago. The strong year-on-year growth was driven by the major platform shifts – accelerated computing and AI. …"

$62.3 billion ✗ not there

It stands in the press release, as the revenue of one quarter. The sentence does not cite that source.

3 It gets one of three results

✓ **Found** VERBATIM

The number stands in a cited source.

✓ **Computed** DERIVED

It is computed from numbers that stand there, and the answer shows how.

✗ **Not found** UNGROUNDED

Neither of the two, so the number is marked.

NOTE The check shows that a number stands in a cited source. It cannot show that it is the right answer for the question. To show this, only a comparison with a **checked answer** (golden answer) can do. **The next page** has such a case, and **Part 3** scores every answer against a checked answer.

Run this in Colab ↗ Sections 2.5 and 2.6 of the notebook run this check as code.

What can go wrong in an answer

Each of these three answers reads fluently and names a source. They differ in who can catch the mistake.

THREE WRONG ANSWERS, AND WHO CAN CATCH EACH

from left to right, harder to catch

1 THE NUMBER CHECK CATCHES IT

An invented computation

recorded · question 9

Q What was NVIDIA's GAAP operating margin for fiscal year 2026?

A … GAAP total revenue for fiscal year 2026 was $215,938 million [1]. Therefore, (**46,107** / 215,938) * 100 = **21.3%** [1].

WHAT THE FILINGS SAY

Operating income **130,387** and revenue **215,938** give a margin of **60.4%**. The answer took 46,107, the operating income of one quarter in its non-GAAP version.

✓ **Caught.** The piece that the answer names holds neither 46,107 nor 21.3%, so the check flags both numbers.

2 ONLY A CHECKED ANSWER CATCHES IT

The wrong period

recorded · question 3

Q Within Data Center, how much **fiscal 2026** revenue came from Compute and how much from Networking?

A For fiscal year 2026 … Compute revenue was **$62.3 billion**, up 75% from a year ago … Networking revenue was a record **$11.0 billion** … [4][5]

WHAT THE FILINGS SAY

12 months — The question asks for the year **$162.4 billion and $31.4 billion**

3 months — The answer gives the fourth quarter **$62.3 billion and $11.0 billion**

✕ **Not caught by the number check**, as every number stands in the named piece. Only a test question with a checked answer shows the mistake.

3 ONLY A PERSON CATCHES IT

An unclear year

an example

Q What was NVIDIA's revenue in 2025, and how much did it grow?

A NVIDIA's revenue in 2025 was **$130,497 million**, up 114% from $60,922 million a year earlier [1].

WHAT THE FILINGS SAY · the fiscal year ends in late January

fiscal 2025	`$130,497 million`
fiscal 2026	`$215,938 million`
calendar 2025	mostly fiscal 2026

✕ **Not caught by any check**, as the number is real and stands in the named piece. Only the person who asked knows which year was meant.

WHAT TO DO

- Ask the model to show every computation, and let code do the arithmetic.
- Let every figure in an answer carry its period, and compare that period with the question.
- Ask which year is meant, or state the reading in the answer: "fiscal year 2026, which ended on 25 January 2026".

Run this in Colab ↗ Exercise 2 of the notebook answers all 14 test questions. Under each answer it lists the numbers and marks every number that has no source.

How much machinery does a QA system need?

A harder question needs more steps, and each added step adds cost and must be verified.

WHAT WE BUILT

level 1 of 3

1 A fixed pipeline

● our design · runs in our notebook

Question → Retrieve → Generate → Numeric grounding → Answer → Audit trail

- Question: as it is asked
- Retrieve: 1 search, six pieces
- Generate: 1 model call
- Numeric grounding: every number is checked
- Answer: with its sources
- Audit trail: kept on record

↻ one more try when a number has no source

Every question takes the same path. This is enough for one fact or one number that stands in one place.

8 of 14
questions answered correctly

1 or 2
model calls for a question

WHAT A HARDER QUESTION WOULD ADD

levels 2 and 3 · orange marks what is added to our design

2 A planned question

beyond this tutorial

A QUESTION THAT NEEDS IT
"What was NVIDIA's Data Center revenue in fiscal year 2026 and in fiscal year 2025?" It has two parts, one for each year.

WHAT IS ADDED
Question → plan → search 2026 → search 2025 → Generate → …

A model first makes a plan: it splits or rewrites the question, and each part gets its own search.

WHAT IT COSTS
One more model call, and one search for each part.

3 Tools and skills

beyond this tutorial

A QUESTION THAT NEEDS IT
"What was NVIDIA's GAAP operating margin for fiscal year 2026?" Two numbers and a division. Our design divided the wrong pair.

WHAT IS ADDED
Question → Generate → tool → result ↻ → …

The model picks a tool (search, read a table, look up the SEC data, calculate), reads the result, and repeats until it has enough. A skill is a written procedure for one kind of question.

WHAT IT COSTS
Several model calls, and nobody knows in advance how many.

Whatever is added, the checks stay. Each sentence names its source, each number is checked, and each answer is kept on record.

Run this in Colab ↗

The notebook implements level 1, including the retry after a failed number check. Levels 2 and 3 are beyond this tutorial.

Exercise 2 · Retrieve, generate and check the numbers

Your task is to do each step yourself on our question about Data Center revenue, the question that our system answered wrongly.

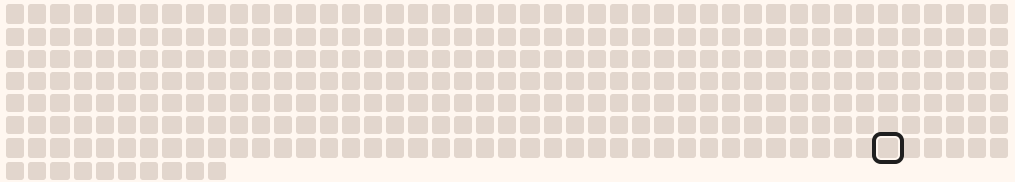
1 **Retrieve** — From the question to six pieces ▸ 2 **Generate** — From six pieces to an answer ▸ 3 **Check the numbers** — From the answer back to its sources

Retrieval searches all the pieces, then re-ranks the candidates. **Click words of the question to search, then pick the candidate that answers it.** show both results

A · SEARCH · 383 pieces → 40 candidates

What was **NVIDIA**'s **Data** **Center** **revenue** in **fiscal** **year** **2026** and in **fiscal** **year** **2025**?

Form 10-K 325 pieces



Press release 32 pieces

CFO commentary 26 pieces

Run the search

A dark square holds all of the words that you clicked, a light one some of them. The framed square is our table.

B · RE-RANK · 40 candidates → 6 pieces

THREE OF THE 40 CANDIDATES · which one answers the question?

Press release · headline of the results — text
"Record quarterly Data Center revenue of $62.3 billion"

10-K · Item 7 · Fiscal Year 2026 Summary — text
"Data Center revenue for fiscal year 2026 was up 68% …"

10-K · Note 16 · revenue by specialized markets — table
Data Center | $193,737 | $115,186 | $47,525 (in millions)

The 40 candidates, in the order of the search



24th · our table

Pick one of the three, and the page shows where the re-ranker placed each of them.

Run this in Colab ↗ Exercise 2 of the notebook answers the 14 questions and prints the report of the number check under each answer.

Ask the documents: your questions, live

live demo

The system we just built, as a website. Open it on your phone and ask it anything about NVIDIA's four filings.

How much did NVIDIA spend repurchasing shares in fiscal 2026? **Ask**

ANSWER

In fiscal year 2026 NVIDIA repurchased **282 million** shares for **$40.4 billion** [1]. As of January 25, 2026 it was authorized to repurchase up to a further **$58.5 billion** of common stock [1].

EVERY NUMBER, CHECKED

VERBATIM 282 million · $40.4 billion · $58.5 billion

[1] 10-K FY2026 › Item 5 › Issuer Purchases of Equity Securities



docai.finhorizons.org/ask

scan it, or type it; the same page runs on the laptop up here

Open the demo ↗

- The notebook's own code, with gpt-oss-120b, OpenAI's open-weight model, called over an API: each answer comes back with its sources in a few seconds. You can also pick another model with your own key.
- **Green** figures are in a cited source word for word, **blue** ones are computed from green ones, **orange** ones are in no source. Tap a citation to read it.
- Try to break it: a figure that is not in the filings, another company, an instruction inside the question, or an as-of date of 1 March 2026.

Part 3: Faithfulness evaluation

- **What to measure**: the pieces, the sources, the numbers and the answer
- **How to score**: a rule, a judge model and a person
- **How to improve**: find the worst answer, fix its cause, score everything again

What is evaluation?

Evaluation asks the same questions again after every change and scores the answers, so that better and worse are measured.

EVALUATION IS A LOOP THAT RUNS AFTER EVERY CHANGE

15 questions with checked answers → the system answers them → **each answer is scored** → the worst answer is read → **one change is made** → all questions are asked again

A score on one day says little. The same questions, asked again after every change, show whether the change helped and whether it broke something else.

WHAT IT NEEDS

A set of questions with answers that a person checked, which page 28 shows.

A way to score an answer, and page 29 compares six of them.

A record of each run, so that two runs can be compared. Our notebook keeps the run of 6 November 2026.

AN ANSWER CAN PASS ONE TEST AND FAIL ANOTHER

Faithful

Is every statement supported by the source that it names?

An answer gives the right net income of fiscal 2025, 72,880, and the paragraph that it names never mentions that figure.

Correct, and not faithful.

Correct

Does the answer agree with the checked answer?

“Net income was 120,067 [1]” names the fiscal 2026 column faithfully, and the question asked about fiscal 2025.

Faithful, and wrong.

Relevant

Does it answer what was asked?

Asked for the operating margin, an answer reports the operating income, 130,387, with a perfect source, and stops.

Faithful and correct, and not an answer.

These three tests fit any question-answering system, so they do not look at numbers. In finance most answers are numbers, so we add two checks: every number must stand in its source, and the system must refuse when the documents do not contain the answer.

Run this in Colab ↗ Block 3 of the notebook scores the answers of Block 2 on the 14 questions that have one checked answer.

What to measure at each step

One score for the whole answer hides where it went wrong, so each step of the pipeline gets a score of its own.

Retrieval
Did the needed piece reach the model?
Hit: the numbers of the checked answer stand in the pieces that were handed over.

Sources
Does each sentence name a source that supports it?
Citation recall: the sentences that name a source. **Citation precision**: the named sources that hold a number of the sentence. **Faithfulness**: the sentences whose numbers all stand in their sources.

Numbers
Does every number have a source?
Numeric grounding: the numbers that stand in a named source or are computed from such numbers.

Answer
Is it right, and does it refuse only when it has to?
Accuracy: the required numbers appear and no forbidden one does. **Abstention**: the fixed refusal, exactly when no answer exists.

ONE EXAMPLE ANSWER, SCORED SIX WAYS BY THE RULE AND BY THE JUDGE — click a score

THE QUESTION · Q01

What was NVIDIA's total revenue for fiscal year 2026, and how much did it grow compared with fiscal year 2025?

AN ANSWER TO SCORE · AN EXAMPLE

S1 Revenue for fiscal year 2026 was **$215,938 million** [1]. S2 That is up **65%** from **$130,497 million** in fiscal year 2025 [1]. S3 Data Center revenue was **$193,737 million** [2]. S4 The growth came from accelerated computing and AI.

[1] 10-K › Item 7 › Fiscal Year 2026 Summary · table (Revenue $215,938 / $130,497 / Up 65%, …)
[2] 10-K › Item 7 › Fiscal Year 2026 Summary · text ("$215.9 billion, up 65%… Data Center revenue … up 68%")

- **Citation recall** — rule 0.75 · judge 0.75
- **Citation precision** — rule 0.67 · judge 0.67
- **Faithfulness** — rule 0.67 · judge 0.50
- **Numeric grounding** — rule 0.75 · judge 0.75
- **Accuracy** — rule 1.00 · judge 1.00
- **Abstention** — rule 1.00 · judge 1.00

Citation recall — Does every sentence cite a source?

BY RULE

Sentences that name at least one source, over all sentences.

```
S1 [1] · S2 [1] · S3 [2]
S4 no citation
3 / 4 → 0.75
```

BY THE JUDGE

The same, counting only sentences that need a source: the judge is asked about each sentence without one (a fact or a figure needs a source, a transition does not).

```
S4 states a fact:
   needs a source
3 / 4 → 0.75
```

> "Here are the figures." needs no source: the rule counts it against the answer, the judge does not. A citation belongs to its own sentence and is never borrowed from the one before (A8).

The rule scores are the notebook's rules run on this answer, and the judge's verdicts are examples that show the arithmetic. Letters in brackets refer to the checklist of page 35.

Run this in Colab ↗

Section 3.3 of the notebook computes the six scores twice: by a rule on the numbers, and by a separate judge model.

The evaluation set

A score is only as good as the questions behind it, and each question needs an answer that a person has checked.

WHAT A SET OF QUESTIONS IS MADE OF

our 15 questions, in notebook/corpus/eval/questions.json

ONE ITEM OF THE SET · Q12

“What was NVIDIA's net income in fiscal year 2025?”

checked answer	Net income for fiscal year 2025 was $72,880 million.
has to contain	$72,880 million
may not contain	$120,067 million, the figure of fiscal 2026
found in	10-K · Consolidated Statements of Income, prior-year column
why it is hard	The fiscal 2026 figure ($120,067 million) sits in the same table; the system must pick the right column.

NINE KINDS OF QUESTION IN OUR 15

a number in a table	3	revenue and its growth
a fact in the text	4	the H20 charge and its cause
a computation	2	the operating margin
two documents	1	GAAP and non-GAAP margin
the press release	1	revenue of the fourth quarter
a footnote, a period	2	regions recast; net income of 2025
no answer in the filings	1	Blackwell units shipped
depends on the date	1	the latest quarterly revenue

THREE WAYS TO WRITE A SET

Written by hand from the filings
Every answer is checked against the source by a person. It is slow, so the set stays small. This is how our 15 were written.

Drafted by a model, checked by a person
A model proposes questions and answers, and a person keeps the correct ones. It is faster, and the questions resemble each other.

Taken from real users
The questions are the ones that are really asked. They need checked answers afterwards, and private questions may not leave the firm.

FIVE STRESS CASES THAT EVERY SET SHOULD HOLD

No answer
q14
The filings never give the number of Blackwell GPUs shipped, so any number is invented.

The wrong period
q12
The figure of fiscal 2026 stands in the same row as the figure that is asked for.

A footnote
q06
The company changed how it counts regions, and only a footnote says so.

The date of the question
q15
The right answer is one number on 1 March and another on 1 September.

A planted instruction
injection
A sentence inside a filing tells the model to ignore its rules. Sources are data and never instructions.

Run this in Colab ↗

Block 3 of the notebook prints the set, and its Section 3.4 plants an instruction in a source and checks that the answer does not follow it.

Six ways to score an answer

Each way is applied to the same recorded answer, which is wrong, and they do not all notice it.

IN OUR NOTEBOOK

A rule on the numbers ●
The required numbers are there or not

A judge model ●
A second model reads the answer

BEYOND OUR NOTEBOOK

Exact match and word overlap
The same words as the checked answer

Vector similarity
The same meaning as the checked answer

An entailment model
Does one text follow from the other

A person
The reference for every other way

● runs in our notebook

A rule on the numbers

RUNS IN OUR NOTEBOOK · REAL RESULT

A rule reads the numbers of the answer in one unit and compares them with the numbers that the checked answer requires.

THE TWO ANSWERS

THE RECORDED ANSWER · WRONG

"For fiscal year 2026, NVIDIA's Data Center revenue was **$62.3 billion**, up 68% from a year ago. [3] For fiscal year 2025, NVIDIA's Data Center revenue was **$38.6 billion**, up 54% from a year ago. [3]"

THE CHECKED ANSWER

"Data Center revenue was **$193,737 million** in fiscal year 2026 and **$115,186 million** in fiscal year 2025, an increase of 68%."

WHAT IT SEES IN OUR ANSWER

The checked answer requires **$193,737 million** and **$115,186 million**. The answer holds $62.3 billion and $38.6 billion, so neither required number appears.

ITS VERDICT

Accuracy 0

The rule is right, and it gives the same result on every run.

ADVANTAGE

It is exact on numbers, it is free, and it never changes its mind.

DISADVANTAGE

It cannot read a sentence without a number, and it needs the required numbers written down for each question.

WHEN TO CHOOSE IT

Choose it wherever the answer is a number.

Run this in Colab ↗

Sections 3.2 and 3.3 of the notebook score every answer by the rule and by a judge model of 7 billion parameters.

What can go wrong in evaluation

A score can mislead as easily as an answer, through the judge model or through the set of questions itself.

THROUGH THE JUDGE MODEL · click a case

1. **It prefers the longer answer**
   Length and position
2. **It does not read the numbers**
   A wrong figure in a good sentence
3. **It has no sense of tolerance**
   60.4% and 60.38%
4. **It judges what sounds right**
   Plausible is not supported
5. **It shares the blind spots**
   The same family of models

A judge tends to choose the longer answer, or the one that is listed first, whatever the two answers say.

HOW IT LOOKS

Two answers to the same question are shown to the judge, and the wordier one wins more often than it should.

WHAT TO DO

Score one answer at a time against its sources, and never ask the judge to choose between two.

THROUGH THE SET AND THE SCORES · click a case

6. **A checked answer is wrong**
   The reference has errors too
7. **Too few questions**
   One question is seven points
8. **Tuned to the questions**
   The set is learnt by heart
9. **Two runs, two scores**
   The same system on another day
10. **A stricter check, a lower score**
    Lower does not mean worse

The answers of the set are written by people, and people make mistakes.

HOW IT LOOKS

While this tutorial was built, several checked answers of our own set were corrected, and the repository lists each correction.

WHAT TO DO

Have a second person review the set, and treat a surprising failure as a reason to read the checked answer again.

FURTHER READING

Bavaresco et al. (ACL 2025), LLMs instead of Human Judges? A Large Scale Empirical Study across 20 NLP Evaluation Tasks.
Kong et al. (ICML 2026), Evaluating LLMs in Finance Requires Explicit Bias Consideration.
Kumar, Sarmah and Pasquali (2026), FinReflectKG - HalluBench: GraphRAG Hallucination Benchmark for Financial Question Answering Systems.

Run this in Colab ↗

Section 3.3 of the notebook lists every item on which the rule and the judge model disagree, with the judge's reason.

Reading the results on 14 questions

Three scores rise with the improved system and two fall, and each fall has a cause that the scores alone do not show.

TWO SYSTEMS ANSWER THE SAME 14 QUESTIONS

run of 6 November 2026 · a 3B generator, a 7B judge, one Colab T4

Baseline

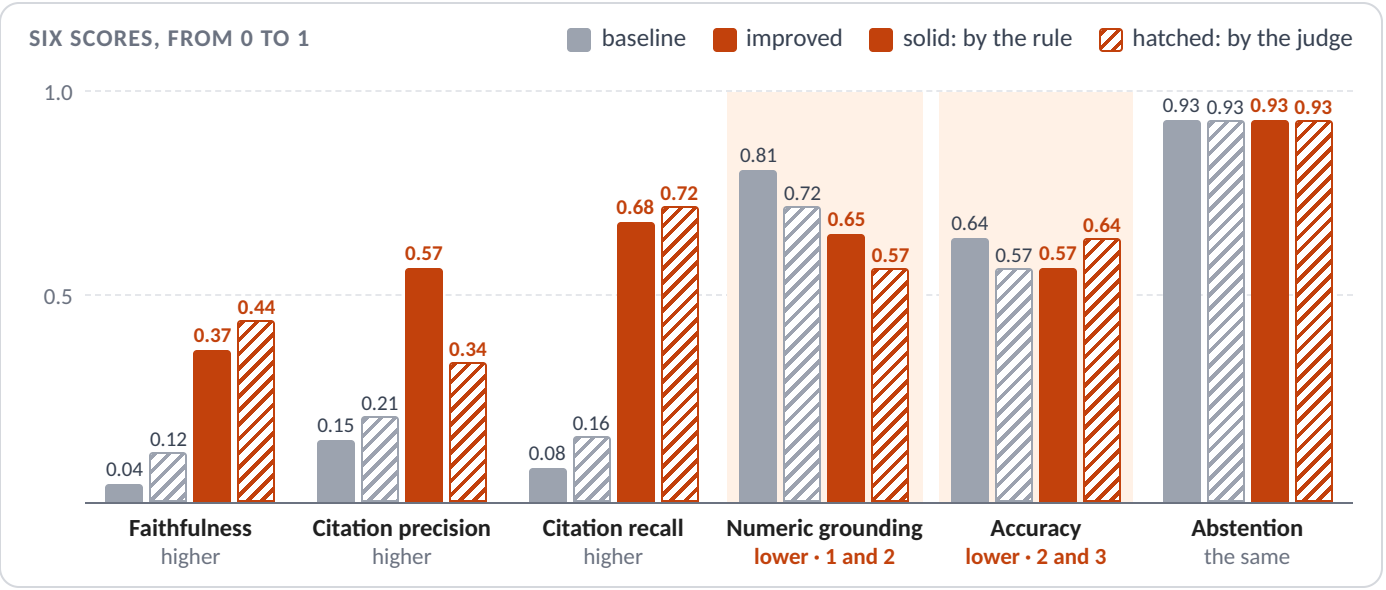
- fixed-window chunks and BM25 only
- a plain “answer the question” prompt
- no grounding check, no retry

Improved

- structure-aware chunks, hybrid retrieval and the re-ranker
- the prompt contract, with citations and the fixed refusal sentence
- the grounding check: if a number fails, the model rewrites its answer once
- the abstention rule

Lesson learned: in our first run, all 8 rewrites gave up and said “not in the documents”. Now giving up does not count: we keep the first answer instead.

THEIR SIX SCORES, AND WHY TWO ARE LOWER FOR THE IMPROVED SYSTEM



1. **Grounding is checked more strictly for it,** so lower here does not mean worse. Each number must be in the source it cites. The baseline hardly cites, so its numbers are checked against all six sources: easier to pass.

2. **It refuses more often.** A refusal on a question that has an answer scores 0 on grounding and accuracy (q12: 72,880 was in source [5]).

3. **The judge disagrees.** It accepts those refusals, so its accuracy goes up (0.57 → 0.64). Same answers, two verdicts: that is why both kinds of bar are shown.

Run this in Colab ↗ Section 3.3 of the notebook prints these scores for both systems, once by the rule and once by the judge.

Explore the answers: every number back to its source

Every recorded answer opens here, and a click on a number or a citation shows the sentence or table row that it comes from.

q01 | q02 | q03 | q04 | q05 | q06 | q07 | q08 | q09 | q10 | q11 | q12 | q13 | q14 | q15 · Mar | q15 · Sep

QUESTION TABLE_LOOKUP

What was NVIDIA's total revenue for fiscal year 2026, and how much did it grow compared with fiscal year 2025?

ANSWER

NVIDIA's total revenue for fiscal year 2026 was **$215.9 billion** [1]. This grew by **65%** compared to fiscal year 2025 [1].

found in a cited source · **computed** from the answer's own figures · **not found**

CHECKED ANSWER

Revenue for fiscal year 2026 was $215,938 million (about $215.9 billion), up 65% from $130,497 million in fiscal year 2025.

HOW THIS ANSWER SCORES

Metric	Rule	Judge	How it is scored
Accuracy	1.00	1.00	Rule: the reference figures are present and no forbidden figure appears; an unanswerable question is declined. Judge: graded CORRECT against the reference answer.
Numeric grounding	1.00	1.00	Rule: each figure is in a cited source after unit conversion, or computed from the answer's own grounded figures. Judge: each figure: stated in the evidence, for the same item

[1] 8-K EX-99.1 · filed 2026-02-25 | [2] 10-K · filed 2026-02-25 | [3] 10-K · filed 2026-02-25 | [4] 10-K · filed 2026-02-25 | [5] 10-K · filed 2026-02-25 | [6] 10-K · filed 2026-02-25

8-K EX-99.1 earnings press release › NVIDIA Announces Financial Results for Fourth Quarter and Fi · text

NVIDIA Announces Financial Results for Fourth Quarter and Fiscal 2026 • Record quarterly revenue of $68.1 billion, up 20% from Q3 and up 73% from a year ago • Record quarterly Data Center revenue of $62.3 billion, up 22% from Q3 and up 75% from a year ago • Record full-year revenue of **$215.9 billion**, up 65% SANTA CLARA, Calif.—Feb. 25, 2026— NVIDIA (NASDAQ: NVDA) today reported record revenue for the fourth quarter ended January 25, 2026, of $68.1 billion, up 20% from the previous quarter and up 73% from a year ago. For fiscal 2026, revenue was $215.9 billion, up 65% from a year ago. For the quarter, GAAP and non-GAAP gross margins were 75.0% and 75.2%, respectively. For fiscal 2026, GAAP and non-GAAP gross margins were 71.1% and 71.3%, respectively. For the quarter, GAAP and non-GAAP earnings per diluted share were $1.76 and $1.62, respectively. For fiscal 2026, GAAP and non-GAAP earnings per diluted share were $4.90 and $4.77, respectively. “Computing demand is growing exponentially — the agentic AI inflection point has arrived. Grace Blackwell with NVLink is the king of inference today — delivering an order-of-magnitude lower cost per token — and Vera Rubin will extend that leadership even further,” said Jensen

Run this in Colab ↗ Section 4.5 of the notebook writes answer_trace.json, the record of every answer that this page shows.

From a score to a fix

A score says that something is wrong, and only a read answer says what, so the worst answer is read before anything is changed.

FIVE STEPS, REPEATED AFTER EVERY CHANGE

1 Rank
The questions are sorted by their score.

2 Read
The worst answer is read in full, with the pieces that the model saw.

3 Name the cause
The failure is assigned to one step of the pipeline.

4 Fix
The smallest change that addresses this cause is made.

5 Score again
All questions are asked again, because a fix can break another answer.

NAME THE CAUSE, WITH THE FAILURES OF OUR OWN RUN

SIX CAUSES, AND THE STEP THAT EACH ONE POINTS TO

The piece never arrived
`q02 in our run`
The table with the answer was ranked 15th, and six pieces are read. The fix is in retrieval.

The wrong period
`q03 in our run`
The answer gave $62.3 billion, a figure of one quarter, for the year. The number check passed it, because the figure is real.

A wrong computation
`q09 in our run`
The model divided 46,107 by 215,938. The number check flagged the input that has no source.

The wrong source is named
`q07 in our run`
The number is right, and the sentence names a source that does not hold it.

A refusal that is wrong
`q12 in our run`
The model said that the documents do not contain the answer, and they do.

An answer that should be a refusal
`q14 passed`
The filings give no number, and the model invents none. Our system refused correctly here.

OUR WORST ITEM, Q12, STEP BY STEP

The question. What was NVIDIA's net income in fiscal year 2025? The system answered: "The documents do not contain this information." The figure, 72,880, stood in one of the six pieces.

The fix that we tried. A hint was added to the question: "check the tables among the sources before saying that the documents lack it". The six pieces stayed the same, and so did the refusal.

The scores afterwards. Every score of the rule stayed at 0. The judge said CORRECT both times, because it believed the refusal, and the required figure overrules the judge.

What it taught us. The hint treated the symptom. The next fix belongs in the prompt or in the model, and after launch the same questions run again after every change.

Run this in Colab ↗ The solution of Exercise 3 in the notebook diagnoses question 12, applies the fix and scores all 14 questions again.

Exercise 3 · Find the worst answer, name its cause, fix it

Your task is to do each step yourself on the 14 answers of our recorded run, from their scores to a fix.

1. **Find the worst answer** — From 14 scores to one answer
2. **Name its cause** — From the answer to the step that failed
3. **Fix it and score again** — From the cause to a new score

The rule gave every answer three scores, and the mean of the three ranks the answers. **Click the answer that you would fix first.**

show the lowest score

	QUESTION	ACCURACY	GROUNDING	FAITHFUL	MEAN
q01	What was NVIDIA's total revenue for fiscal year 202…	1	1	1	**1.00**
q02	What was NVIDIA's Data Center revenue in fiscal ye…	0	0.25	0	**0.08**
q03	Within Data Center, how much fiscal 2026 revenue …	0	1	0.5	**0.50**
q04	What charge did NVIDIA record in fiscal 2026 in con…	1	1	1	**1.00**
q05	How concentrated was NVIDIA's revenue in fiscal 20…	1	0.5	0	**0.50**
q06	What percentage of NVIDIA's fiscal 2026 revenue ca…	1	1	0.33	**0.78**
q07	How much did NVIDIA spend repurchasing shares in…	1	0.33	0	**0.44**
q08	What were NVIDIA's GAAP and non-GAAP gross mar…	0	1	0	**0.33**
q09	What was NVIDIA's GAAP operating margin for fisca…	0	0	0	**0.00**
q10	By how much did NVIDIA's inventories increase bet…	0	1	0.33	**0.44**
q11	What were NVIDIA's revenue and Data Center reven…	1	1	1	**1.00**
q12	What was NVIDIA's net income in fiscal year 2025?	0	0	0	**0.00**
q13	How many employees did NVIDIA have at the end o…	1	0	0	**0.33**
q14	How many Blackwell GPUs did NVIDIA ship in fiscal …	1	1	1	**1.00**

Which answer would you fix first?

The table shows the 14 questions with three of their scores, as the rule computed them. A score of 0 is marked.

Click a question to read its recorded answer and to see how many answers score lower.

Run this in Colab ↗

Exercise 3 of the notebook scores the 14 answers, opens the worst one and scores everything again after a fix.

What we learned, as a checklist for your own system

Every principle here was broken at least once while we built this pipeline. Click a group to see how.

Twenty principles | **All 250 checks, on the website ↗**

Know what a number is 4
kind, scale, precision; is it a number at all?

Signs, sums and stray digits 3
bracketed outflows, computed figures, years, notes

Read tables as tables 1
spans, groups, units and footnotes decide a cell

Same evidence, same rule, everywhere 3
every check sees the same passages and rules

Define exactly what you measure 4
refusals, citation metrics, gold answers, fair tests

Guard automatic fixes 1
a better score must not buy a worse answer

Test the evaluator, record everything 4
the judge, the rules and the runs can all be wrong

A1 Compare a number only with numbers of the same kind: percentages, money, counts, per-share amounts and durations.

Without it: An answer's "operating margin was 4.9%" counted as supported because the cited table contained EPS "$4.90".

A3 Normalise scale and notation before comparing: $215.9 billion, 215,938 (in millions) and 215938000000 are one number.

Without it: Comparing strings treats one number as several different numbers.

A4 Set the tolerance from the precision of the number being checked, symmetric on both sides, and match whole numbers only.

Without it: Through floating-point error 60.2% passed and 60.6% failed against 60.4 ± 0.2; "22%" matched inside "122%".

A19 Know where numbers live and what is a number: row labels hold numbers, dates are not numbers, "per share" in a label does not make a per-share amount, a table's unit can sit in the sentence before it, and numbers in a picture are invisible to a text parser.

Without it: A bond's 3.20% in its row label was never read; 12/10/2025 was split into 12 and 10 and supported "$12 million"; "Weighted-average shares used in per-share computation" is a share count;

The whole checklist is on the tutorial website.

Part 4: Production Concerns



- **Concerns in backtest or review a past decision**: what information did I have?
- **The speed of library update**: is the new filing in the system yet?
- **Cost and latency**: how long do I wait, and what does it cost?
- **Hallucinated Number**: where did each number come from? (Audit Trail)

Backtest or review: what information did I have?

- You backtest an idea, or someone asks why you bought in March. Either way you need the numbers **you could see on that day**, not the ones that came out later.
- Companies change their numbers afterwards: **corrections** (a 10-K/A), **restatements**, and later reports that publish earlier quarters again under **new categories**.
- A chatbot gives today's version. In a backtest that is **cheating without knowing it** (look-ahead bias): the idea looks better than it really was.

The fix: every question carries a date. The system **reads only what was public by then**, in its latest version, and records the date so the answer can be rerun.

Portfolio manager · backtesting 1 March 2026

What was NVIDIA's most recently reported quarterly revenue?

WITHOUT THE FIX the system reads everything in its library

$96,221 million for Q2 FY2027. Source: the 10-Q filed 26 August 2026.

✕ **Filed almost six months after 1 March: look-ahead bias.**

WITH THE FIX the system reads only what was filed by 1 March

$68,127 million for Q4 FY2026. Source: the CFO commentary filed 25 February 2026.

✓ **The latest report that was public on 1 March.**

Run this in Colab ↗ notebook 4.1 · as_of_date, q15 at two dates

The speed of library update

- In earnings season, dozens of the companies you follow report in the same week. You want to ask about a filing **the day it lands**, not a week later.
- If the new filing is not in the system yet, you ask for the latest revenue and get **last quarter's number**, with no warning that it is out of date.
- Adding new filings may break the **citation record in your old answers**.

The fix: a new filing is **added on its own**, in seconds, and nothing else is rebuilt. Each citation is tied to **the text itself**, not to its position in the library, so it cannot shift when filings are added.

THE LIBRARY ON 26 AUGUST 2026

Form 10-K, fiscal 2026	filed 25 Feb 2026	not touched
Q4 FY2026 press release	filed 25 Feb 2026	not touched
Q4 FY2026 CFO commentary	filed 25 Feb 2026	not touched
Form 10-Q, Q2 fiscal 2027	filed 26 Aug 2026	added in 2.4 s

WHAT YOU CAN RELY ON

2.4 seconds
from the filing arriving to ready to ask

383 of 383
earlier paragraphs and tables left exactly as they were

Run this in Colab ↗ notebook 4.1 · the 10-Q is added to the index without a rebuild

How long do I wait, and what does it cost?

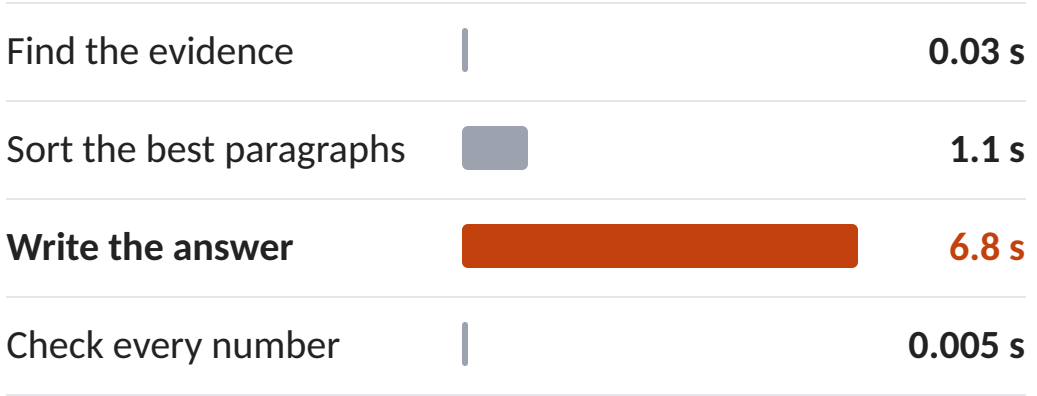
- While you listen to an earnings call, you want a quick check of a specific number, so the answer must come **in seconds**. On a busy morning you ask dozens of questions, so we need to make sure the token bill is under control.
- A system that is not carefully designed for latency often takes **a minute or two** to answer, because it has many filings to search and the question passes through a long chain of steps.
- And the token bill grows fast, because you pay for **every word the model reads** on every question.

The fix: small, cheap models find the evidence, and the large model reads only **a few good paragraphs**. Each filing is read once and the result kept. Quality is checked afterwards, never while you wait.

Run this in Colab ↗ notebook 4.2 · time_stages

ONE QUESTION, START TO FINISH

about 8 seconds



Timed on a basic cloud GPU (Colab T4), 6 November 2026, with a small model and four filings. Reading a new 10-K takes another 0.7 s, once.

Hallucination: where did each number come from?

- A model can give a **hallucinated** number:
  Made up: it does not exist in the filing at all.
  Wrong period: it is from another quarter or year.
  Misread: it is real, but means something else.
- Before you use a number, you want to **fact-check** it:
  Accuracy: does it match verified records?
  Context: is it the right period, place or group?
  Source: where is it from, so you can cite it?
- Without a record of what the model read, the only way to check a number is to **look it up in the filing yourself**.

The fix: every answer is saved with a **receipt**: what was asked and when, the exact text the model read, which version of the system answered, and the checks it passed. A receipt is never edited afterwards.

THE ANSWER YOU GET

Fourth-quarter revenue was a record **$68.1 billion**, up **73%** from a year ago and up **20%** sequentially. [1] Fourth-quarter Data Center revenue was a record **$62.3 billion**, up **75%** from a year ago and up **22%** sequentially. [2]

WHERE EACH NUMBER COMES FROM

> [1] **Q4 FY2026 press release, filed 25 Feb 2026**
> "Record quarterly revenue of **$68.1 billion**, up **20%** from Q3 and up **73%** from a year ago"

> [2] **Same press release, Data Center section**
> "Fourth-quarter revenue was a record **$62.3 billion**, up **22%** from the previous quarter and up **75%** from a year ago"

✓ **6 of 6 numbers found word for word in the text the model read**

Saved with it: when it was asked, and which version of the system answered.

Run this in Colab ↗ notebook 4.3 · AuditRecord, audit_log.jsonl

Responsible use: the rules and the weak spots

The rules: what compliance will ask

- **Which rules apply?** It depends on where you work:
  US: the bank regulators' guidance does not cover generative AI. Each firm sets its own controls for its AI tools [1].
  EU: the AI Act already bans practices such as social scoring, and people must be told when they are dealing with AI. Rules for high-risk uses such as credit scoring start on 2 December 2027 [2].
  UK: the financial regulator has made no AI-specific rules. Its existing rules apply to firms' use of AI [3].
- **Who is accountable?** You are. The system writes a draft, and a person must review and approve it [4].
- **May we use these documents?** Filings are public. Licences for call transcripts and broker research may forbid AI use.
- **Who can see what?** Each person gets answers only from documents they are allowed to open.

The weak spots: what to check yourself

- **When an answer combines numbers from several reports**, check where each number came from.
  Possible fix: show a source next to each number.
- **When a number is read off a chart or a figure**, check it yourself. The error rates there are not yet low enough to trust.
  Possible fix: cross-check it against the filing's tables.
- **Do not rely on the system's own test scores.** In finance these scores are easily inflated, for example by look-ahead bias [5].
  Possible fix: use test questions with human-checked answers.
- **Do not assume that a model with extra training is safer.** In one study, fine-tuning raised the share of made-up numbers from 5% to over 80% [6]. In another, about one citation in seven was attached afterwards to justify the answer [7].
  Possible fix: check every number, whatever the model.

[1] Board of Governors of the Federal Reserve System, FDIC and OCC. *Supervisory Guidance on Model Risk Management.* SR Letter 26-2, 17 April 2026 (replaces SR 11-7).
[2] *Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, as amended by Regulation (EU) 2026/1744 (Digital Omnibus on AI), OJ L, 24 July 2026.
[3] Financial Conduct Authority. *AI in financial services: good practice and areas for improvement.* 19 October 2026.
[4] Financial Stability Board. *Sound Practices for Responsible Adoption of Artificial Intelligence (AI): Final report.* 15 October 2026.
[5] Y. Kong, H. Lee, Y. Hwang, A. Lopez-Lira, B. Levy, D. Mehta, Q. Wen, C. Choi, Y. Lee, S. Zohren. *Evaluating LLMs in Finance Requires Explicit Bias Consideration.* ICML 2026. arXiv:2602.14233.
[6] X. Li, P. Liu. *When Financial Fine-tuning Fails: A Three-Level Detectability Analysis of Numerical Hallucination in Domain-Adapted Language Models.* arXiv:2609.04806, 2026.
[7] M. Khan, M. S. I. Bhuyan. *Attributable Post-Rationalization in RAG Citations: A Controlled Reproduction and an RLVR Comparison.* arXiv:2609.23053, 2026.

Which third-party tools can my team use directly?

What the system needs	Third-party tools you can use directly
Reading the filings turns each filing into clean paragraphs and tables	Docling · Unstructured · MinerU · Azure Document Intelligence · Amazon Textract · Google Document AI
Search finds paragraphs by keyword and by meaning, filtered by date	Elasticsearch · OpenSearch · Vespa · Qdrant · Weaviate · Milvus · pgvector · Pinecone · Azure AI Search · Google Agent Search
Sorting the results puts the most relevant paragraphs first	Cohere Rerank · Voyage AI · Jina Reranker · Text Embeddings Inference
The model that writes the answer run an open model yourself, or call one from your cloud	vLLM · Text Generation Inference · SGLang · Amazon Bedrock · Gemini Enterprise Agent Platform · Microsoft Foundry
Instructions and run history keeps every version of the instructions and records each run	Langfuse · LangSmith · MLflow · Braintrust · Weave · Arize Phoenix · Helicone · PromptLayer
Testing scores the system on a fixed set of checked questions	promptfoo · DeepEval · Ragas · Inspect
The receipt log stores each answer's receipt so that it cannot be edited	Amazon S3 Object Lock · Azure immutable Blob Storage · Google Cloud Bucket Lock

These are examples, not recommendations, and the order means nothing. Pick what fits your team. Click any name to open its website.

Run this in Colab ↗ notebook 4.4 · a search-engine index sketch and a prompt registry

Back to the first question

We opened with an answer that reads well, and we close with the same question.

ONE QUESTION, TWO ANSWERS

THE GOOD ANSWER OF PAGE 2

"NVIDIA's Data Center revenue for fiscal 2026 was a record $193.7 billion, up 68% year over year."

It is right, and a reader has no way to know that. Its number comes from a summary on the web, and nothing of it is saved.

OUR OWN SYSTEM, ON THE SAME QUESTION

"For fiscal year 2026, NVIDIA's Data Center revenue was $62.3 billion, up 68% from a year ago. [3]"

It is **wrong**. The checked answer is $193,737 million, and $62.3 billion is the revenue of one quarter.

WHAT THE SYSTEM SHOWED ABOUT ITS OWN ANSWER

Caught	3 of its 4 numbers have no source.
Scored	Accuracy 0 against the checked answer.
Explained	The table with the answer was ranked 15th, and six pieces are read.
Recorded	The six pieces, the prompt and the checks are saved.

THE FOUR THINGS THAT PAGE 2 ASKED FOR, AND WHERE EACH ONE WAS BUILT TODAY

1 Proof

Every number of an answer is looked up in the source that its sentence names.

Part 2 · numeric grounding

2 Context

A number keeps its table, its row, its column and its unit from the filing to the answer.

Part 1 · parsing and chunking

3 Date

Every piece carries its filing date, and a question about the past reads only what existed then.

Parts 1 and 2, and Part 4

4 Record

Every answer is saved with what the model read, and the same questions are asked again after each change.

Parts 2 and 3

A trustworthy system is not one that is never wrong. Its wrong answers are caught, explained and fixed.

Where the effort pays off first

In our experience the steps before the model decide more than the model does, and the checks decide what can be trusted.

EIGHT DESIGN CHOICES, IN THE ORDER OF THEIR IMPACT

CHOICE	IMPACT	WHY	SHOWN ON
Parser and table handling	HIGH	Every number that an answer cites has passed through the parser.	pages 7–8
Metadata on every piece	HIGH	Without it there are no sources to name, no date filter and no record.	page 9
The number check	HIGH FOR TRUST	It does not make answers more accurate. It makes a wrong number visible.	page 20
The judge model	HIGH FOR SCORES	A judge that is biased spoils every score that it gives.	pages 29–30
The re-ranker	MEDIUM TO HIGH	The six pieces at the top are all that the model reads.	page 16
Keyword and vector search	MEDIUM	Each one misses what the other finds, so both are searched.	page 13
The size of the model	MEDIUM	It matters little once the right pieces arrive.	page 22
The rules of the prompt	MEDIUM	They make an answer measurable, and they do not make it accurate.	page 19

TODAY WAS ABOUT DOCUMENTS · WHAT WE DID NOT COVER

Sources other than documents
Numbers that already stand in a database are better queried than searched, for example with SQL. A query needs a written definition of each table, column and metric.

Pictures, charts and sound
Our filings arrive as web pages. What a model reads from a scan, a chart or a recorded call needs a check of its own.

Many companies
Our library holds one company. With many, the company and the period have to be fixed before the search.

Planned questions, tools and skills
Page 22 describes them, and our notebook does not run them, so we report no gain for them.

A larger model and more questions
The run uses a model of 3 billion parameters and 14 questions. Both are small on purpose, so that it runs on a free Colab.

COMING SOON We are preparing **a comprehensive guide to QA systems in finance**. It will cover what we did not cover today. **Stay tuned.**

What changed in 2025–2026 click a paper

Where each block's method is heading, and the papers behind it. Click a paper for a one-line summary and the link.

1 Parsing	Parsers of 1–4B parameters now match far larger ones; finance-specific benchmarks show how far public parsers are from deployment.	FinixDoc · LightOnOCR · GLM-OCR · Table QA survey · Structure-aware chunking
2 Retrieval	Pin down the company, the fiscal year and the section first, then collect evidence; re-rank by entity, metric, period and numeric span.	HC-RAG · FinSAgent · LOFin · FinCARDS · Retrieval failures
2 Grounding	Split answers into typed atomic claims; recompute derived figures against the table; do the arithmetic on a fact ledger, not in the model.	FinGround · Fact ledgers · FAITH
2 Agents	A split question finds more evidence when a question has several parts, and an agent costs more than a fixed pipeline, so its budget and its rule for stopping matter.	Question decomposition · Is agentic RAG worth it? · Budgets for agentic search · MimirRAG
3 Evaluation	Verify citations at the source; one in seven is written after the answer; validate a judge for each task; fine-tuning worsens numeric errors.	CiteGuard · VeriCite · Khan and Bhuyan · Bavaresco et al. · Li and Liu · Kong et al. (ICML)
4 Production	Internal-state probes as cheap triage; chunker gains that fail a cost check; point-in-time models, look-ahead bias and temporal leakage audited.	Confidently Wrong · Chunking cost study · HindsightBench · Temporal leakage · Look-Ahead-Bench

FinixDoc: Rethinking Financial Document Parsing Beyond Saturated Benchmarks

Wang et al. · arXiv 2026

Agentic parsing system with a 4B Qwen3-VL-based parser built for financial documents, and a benchmark that exposes the gap between leaderboard scores and deployment accuracy.

Open the paper ↗

For full citations, see the resource hub.

Your questions click a question



Four questions to start with

- Can I use this on another company, or on my own documents?
- Why not put the whole 10-K into a long-context model?
- Won’t a bigger model fix this?
- Can I trust an LLM as the judge?

…and anything else, now.

About today, your own documents, or something that broke in your notebook.

Take it home

Notebook in Colab ↗
No GPU needed: it replays our recorded run.

Slides on GitHub ↗
This deck as HTML, PowerPoint and PDF.

Ask the documents ↗
An early question-answering site on NVIDIA’s filings.

TWO OPEN-SOURCE PROJECTS

Evaluation checklist ↗
The failure cases to test for when you evaluate: 250 checks, by pipeline stage.

Resource hub ↗
The papers we found, their methods, and which step of this pipeline each could replace.

All of it is linked from
docai.finhorizons.org

FinHorizons Foundation

An open research community for responsible AI in finance.
Every Saturday on Zoom we hold either a seminar or a reading group.

Seminar

Researchers and practitioners talk about AI in finance. Every talk is recorded, and the slides stay online.

or

Reading group

One topic per session. Several people present recent papers on it, then we discuss them together.

NEXT **Saturday 21 November** · Reading group on **world models** · with Qingren Yao and Yudi Zhang

THE LAST FOUR TALKS

7 NOV
Steven Wang
Engineering the Investment Brain

31 OCT
Yuxuan Wang
Transaction Fraud Defense in the Wild

24 OCT
Yuqi Nie
Financial Regime Identification and Regime-Aware Factor Allocation

17 OCT
Yoontae Hwang
Decision Focused Learning for Portfolio Optimization

Stay in touch

Website
finhorizons.org

Contact **events@finhorizons.org**

Thank you · All materials are open source and remain available after the conference.

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